

Anthony Qi

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CS junior with experience building compilers, systems software, and backend-focused projects in C, Rust, and OCaml.

EDUCATION

University of Maryland, College Park | *Expected Graduation: May 2028*

B.S. in Computer Science & Immersive Media Design (Computing)

GPA: 3.5/4.0 | **Award:** **Presidential Scholarship Recipient** (4 years)

Relevant Coursework: Systems Programming · Organization of Programming Languages · Algorithms · Discrete Structures · Object Oriented Programming · Linear Algebra · Data Structures and Algorithms

TECHNICAL SKILLS

Languages: Java, C, OCaml, Rust, Python, JavaScript, C#, SQL, x86-64 Assembly, Typescript

Other Skills: Git, Bash, Linux/Unix, GDB, Valgrind, JUnit, MySQL

Concepts: Compilers, Systems Programming, Multithreading, Memory Management, Garbage Collection, Static Type Systems, Data Structures & Algorithms, Machine Learning, NLP Model Training (Intent Classification)

PROJECTS

Personal Portfolio Website (React, Typescript, Vite) | *Summer 2026*

- Rebuilt personal portfolio site from vanilla HTML/JS/CSS into a React, Typescript, Vite, and Tailwind CSS deployed via Github Actions to Github Pages.

Rust Garbage Collection (Rust) | *Spring 2026*

- Implemented three garbage collectors in Rust: reference counting, mark-and-sweep, and stop-and-copy, including recursive heap traversal and live-object compaction.

OCaml C Compiler (OCaml) | *Spring 2026*

- Built a full compiler pipeline in OCaml: lexer, parser, AST evaluator, static type checker, and stack-machine code generator for a SmallC-like language.

Unix Shell (C) | *Fall 2025*

- Built a Unix-style shell from scratch supporting fork/exec, I/O redirection, pipelines, and background processes via wait() and signal handling.

Explicit List (EL) Malloc and Matrix Operation Optimization (C) | *Fall 2025*

- Implemented an explicit free list allocator using mmap, with splitting and coalescing for dynamic heap management.
- Optimized multithreaded matrix multiplication using pthreads, mutex synchronization, and cache-aware blocking to improve memory locality and performance.

Minecraft Fabric Mods (Java) | *Jun. – Sep. 2025*

- Built two Minecraft Fabric mods featuring server-side state tracking (boss bar + commands) and custom gameplay mechanics (enchancements, cooldown systems, and textured items).

EXPERIENCE

MascotGo | Remote | *Jan. 2026 – Present* | *Student In Residence*

- Fine-tuning a lightweight NLP model using Unsloth on a dataset of 11,000+ labeled queries for intent classification across student support categories.
- Collaborating with a cross-functional team to clean and standardize training data, validate model accuracy, and route classified queries to appropriate response tiers.

National High School Game Academy | Carnegie Mellon University | *Jul. – Aug. 2023* | *Student Game Developer*

- Built a fully playable game prototype in Unity/C# with a multidisciplinary team, integrating original art and audio under sprint-based deadlines.
- Presented the final game to CMU faculty and industry professionals at a public showcase.

ROAR Club | Valley Christian High School | *Jan. 2023 – Jun. 2024* | *Co-Founder & VP*

- Co-founded a machine learning club focused on autonomous racing; taught ML concepts through presentations to 20+ members.

Home Depot | San Jose | *Jun. 2025 – Aug. 2025* | *Service Specialist*

- Handled customer inquiries, returns, and equipment rentals (trucks, vans, carpet cleaners) in a high-volume retail environment.

ADDITIONAL

Languages: English (native), Chinese (fluent), Japanese (reading), Spanish (basic) | **Creative:** Drawing, 3D Modeling, Digital Media